CSE STUDY HUB

Generative AI & Prompt Engineering

UNIT 6 · Research Component

What this unit is about

This unit builds understanding of **Research Component**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Choose a focused research question and define the problem

2. Conduct a literature review and identify a research gap

3. Select models/tools and define comparison criteria

4. Design a reproducible experiment or case study

5. Metrics: accuracy, relevance, latency, cost and human ratings

6. Record prompts, versions, datasets and evaluation procedure

7. Analyse findings, limitations, ethics and threats to validity

8. Write a concise report and propose future work

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 6 NOTES · PART 1

Choose a focused research question and define the problem

Choose a focused research question and define the problem is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Conduct a literature review and identify a research gap

Conduct a literature review and identify a research gap is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Select models/tools and define comparison criteria

Compare the alternatives in Select models/tools and define comparison criteria using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Research Component, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 6 NOTES · PART 2

Design a reproducible experiment or case study

Design a reproducible experiment or case study is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Metrics: accuracy, relevance, latency, cost and human ratings

Metrics: accuracy, relevance, latency, cost and human ratings is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Record prompts, versions, datasets and evaluation procedure

A strong prompt specifies the task, context, constraints and desired output format. Add examples when useful, break complex tasks into steps, and verify facts and assumptions in the response.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Research Component, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 6 NOTES · PART 3

Analyse findings, limitations, ethics and threats to validity

Compare the alternatives in Analyse findings, limitations, ethics and threats to validity using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Write a concise report and propose future work

Write a concise report and propose future work is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Research Component, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Choose a focused research question and define the problem** — know the definition, core working and one example.

• **Conduct a literature review and identify a research gap** — know the definition, core working and one example.

• **Select models/tools and define comparison criteria** — know the definition, core working and one example.

• **Design a reproducible experiment or case study** — know the definition, core working and one example.

• **Metrics: accuracy, relevance, latency, cost and human ratings** — know the definition, core working and one example.

• **Record prompts, versions, datasets and evaluation procedure** — know the definition, core working and one example.

• **Analyse findings, limitations, ethics and threats to validity** — know the definition, core working and one example.

• **Write a concise report and propose future work** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Research Component. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Choose a focused research question and define the problem and explain its importance.

Q2. Explain the working/steps involved in Conduct a literature review and identify a research gap.

Q3. Differentiate between two related concepts from Research Component.

Q4. Give one practical application of Write a concise report and propose future work and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Research Component.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.